

Decision Making with Business Statistics

Week 4: Project: E-News Express

Define Problem Statement and Objectives

Import all the necessary libraries

```
In [ ]: # Installing the libraries with the specified version.
!pip install numpy==1.25.2 pandas==1.5.3 matplotlib==3.7.1 seaborn==0.13.1
scipy==1.11.4 -q --user
```

```
[?25l [90m [0m
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```

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[?25h Installing build dependencies ... [?25l[?25hdone
[1;31merror[0m: [1msubprocess-exited-with-error[0m

[31m[0m [32mGetting requirements to build wheel[0m did not run
successfully.
[31m[0m exit code: [1;36m1[0m
[31m[0m See above for output.

[1;35mnote[0m: This error originates from a subprocess, and is
likely not a problem with pip.
Getting requirements to build wheel ... [?25l[?25herror
[1;31merror[0m: [1msubprocess-exited-with-error[0m

[31m[0m [32mGetting requirements to build wheel[0m did not run
successfully.
[31m[0m exit code: [1;36m1[0m
[31m[0m See above for output.

[1;35mnote[0m: This error originates from a subprocess, and is likely
not a problem with pip.

```

Note: After running the above cell, kindly restart the notebook kernel and run all cells sequentially from the start again.

```

In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from scipy.stats import norm

```

Reading the Data into a DataFrame

```

In [4]: from google.colab import drive
drive.mount('/content/drive')

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```

In [5]: df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/abtest.csv')

```

Explore the dataset and extract insights using Exploratory Data Analysis

- Data Overview
- Viewing the first and last few rows of the dataset
- Checking the shape of the dataset
- Getting the statistical summary for the variables
- Check for missing values
- Check for duplicates

In [6]: `df.head(5)`

Out[6]:

	user_id	group	landing_page	time_spent_on_the_page	converted	lang
0	546592	control	old	3.48	no	Span
1	546468	treatment	new	7.13	yes	Engl
2	546462	treatment	new	4.40	no	Span
3	546567	control	old	3.02	no	Fren
4	546459	treatment	new	4.75	yes	Span

In [7]: `df.tail(5)`

Out[7]:

	user_id	group	landing_page	time_spent_on_the_page	converted	lan
95	546446	treatment	new	5.15	no	Span
96	546544	control	old	6.52	yes	Engl
97	546472	treatment	new	7.07	yes	Span
98	546481	treatment	new	6.20	yes	Span
99	546483	treatment	new	5.86	yes	Engl

In [8]: `df.shape`

Out[8]: (100, 6)

In [9]: `df.describe()`

Out[9]:

	user_id	time_spent_on_the_page
count	100.000000	100.000000
mean	546517.000000	5.377800
std	52.295779	2.378166
min	546443.000000	0.190000
25%	546467.750000	3.880000
50%	546492.500000	5.415000
75%	546567.250000	7.022500
max	546592.000000	10.710000

```
In [10]: #there are no missing values
df.isnull().sum()
```

Out[10]:

	0
user_id	0
group	0
landing_page	0
time_spent_on_the_page	0
converted	0
language_preferred	0

dtype: int64

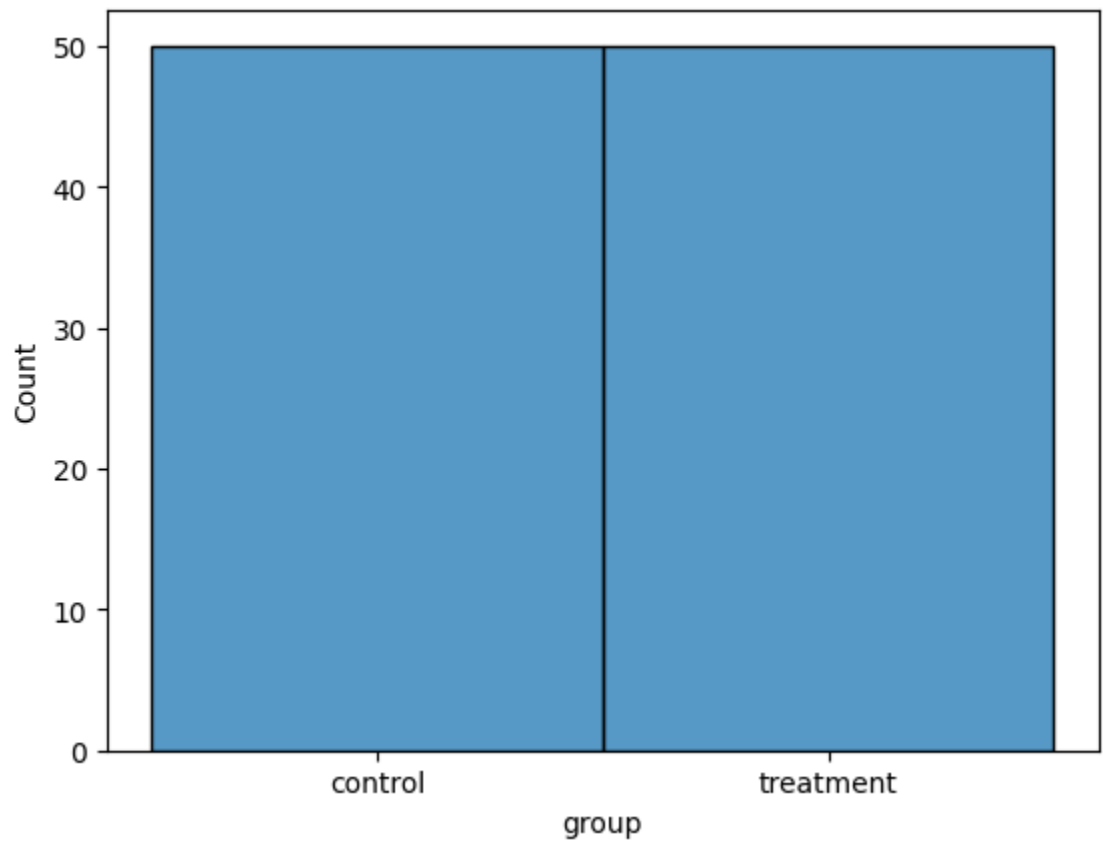
```
In [11]: #no duplicates either
df.duplicated().sum()
```

Out[11]: np.int64(0)

Univariate Analysis

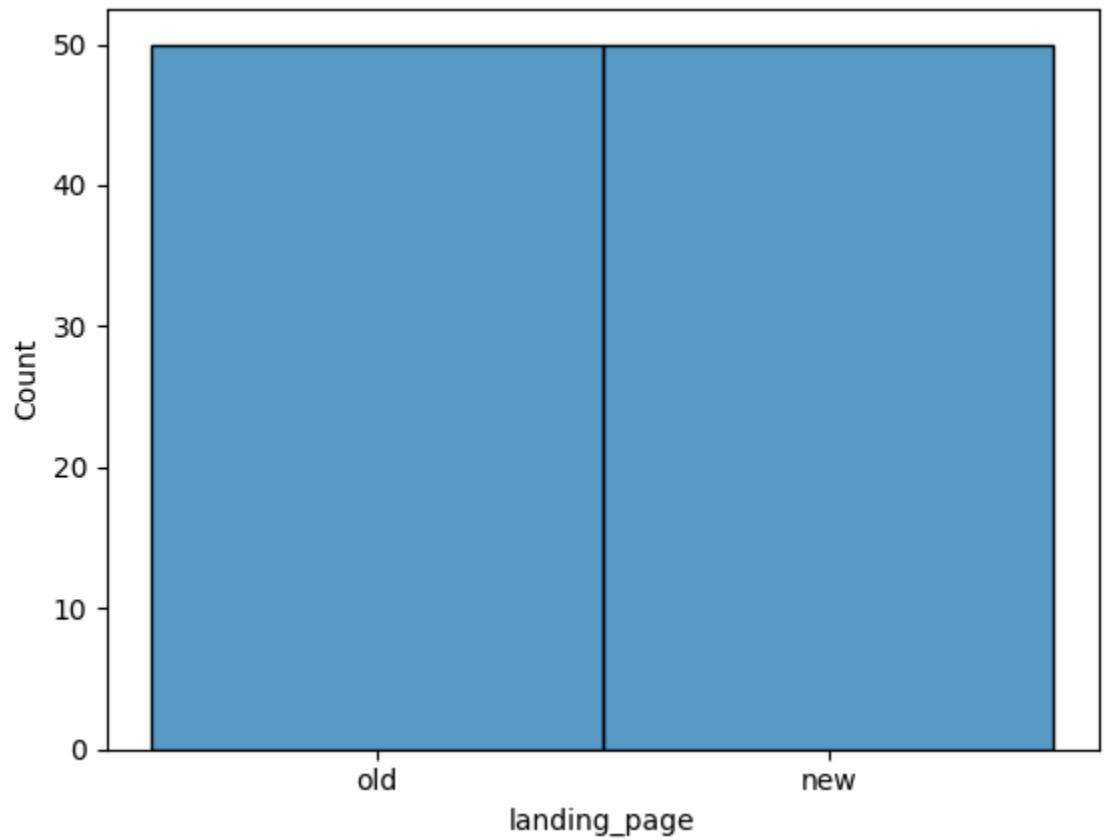
```
In [12]: sns.histplot(df['group'])
```

Out[12]: <Axes: xlabel='group', ylabel='Count'>



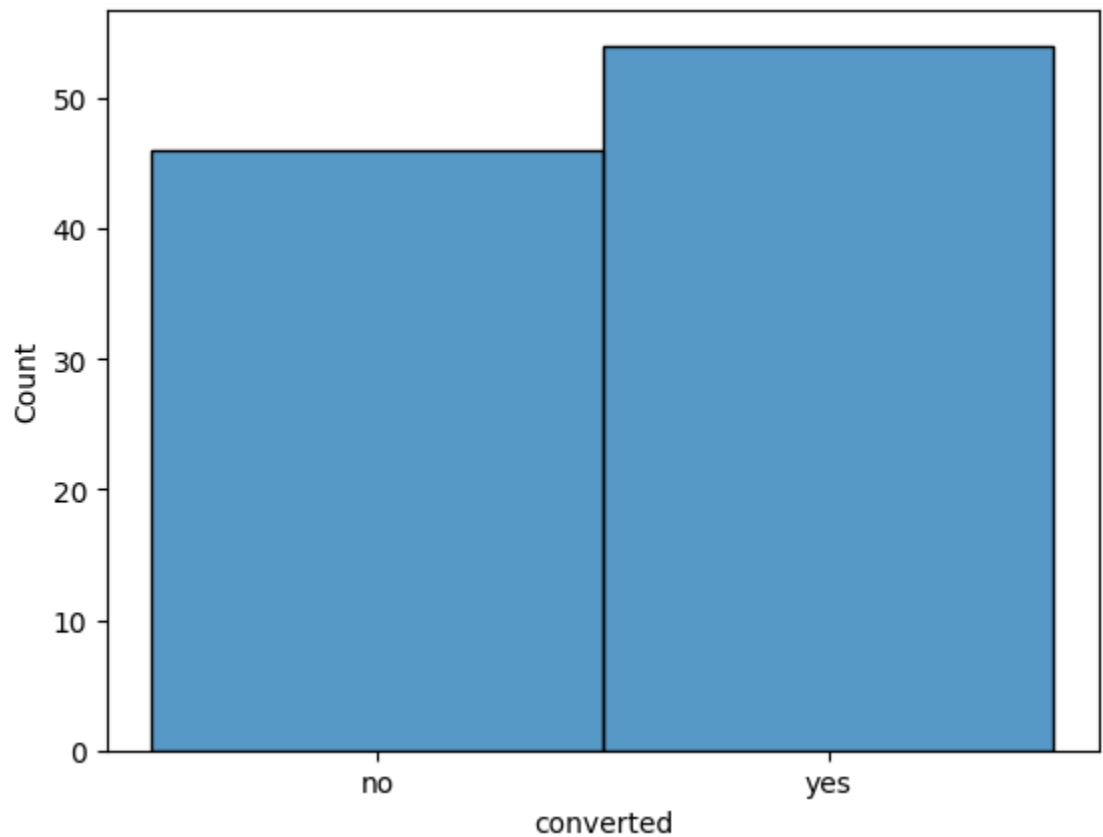
```
In [13]: #same as group column  
sns.histplot(df['landing_page'])
```

```
Out[13]: <Axes: xlabel='landing_page', ylabel='Count'>
```



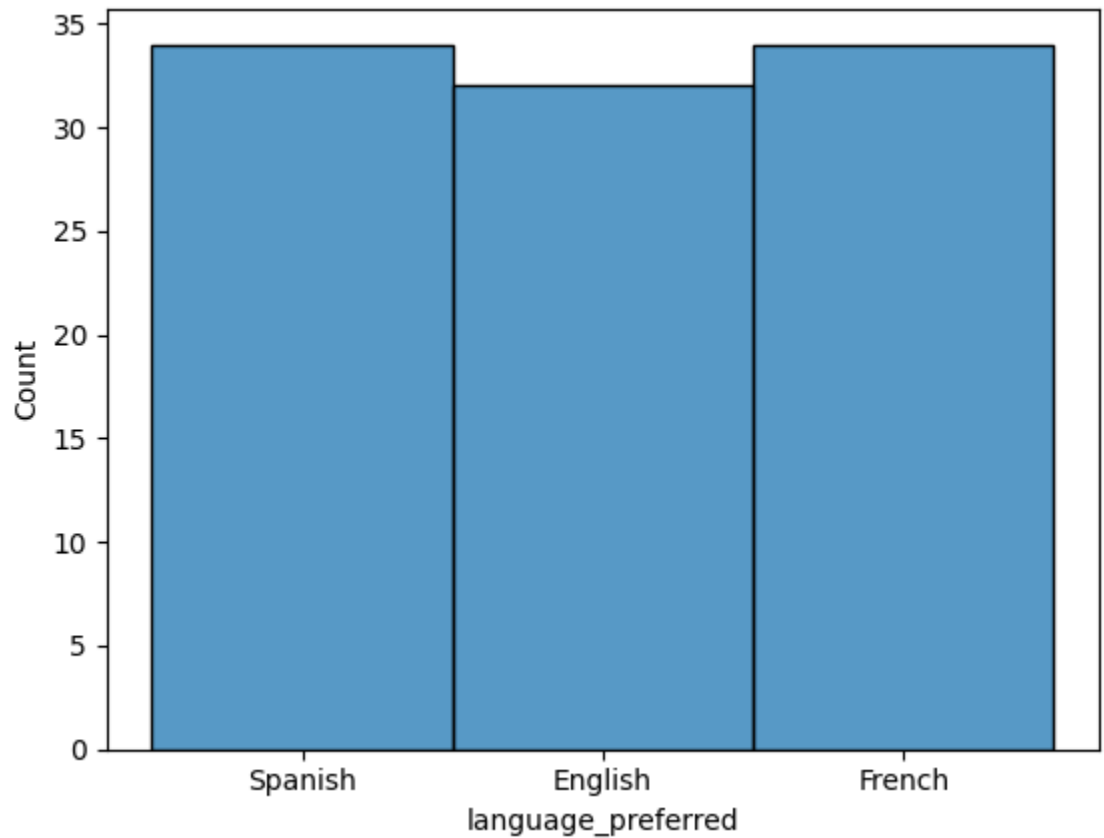
```
In [14]: #45 not converted, 55 were converted to switch  
sns.histplot(df['converted'])
```

```
Out[14]: <Axes: xlabel='converted', ylabel='Count'>
```



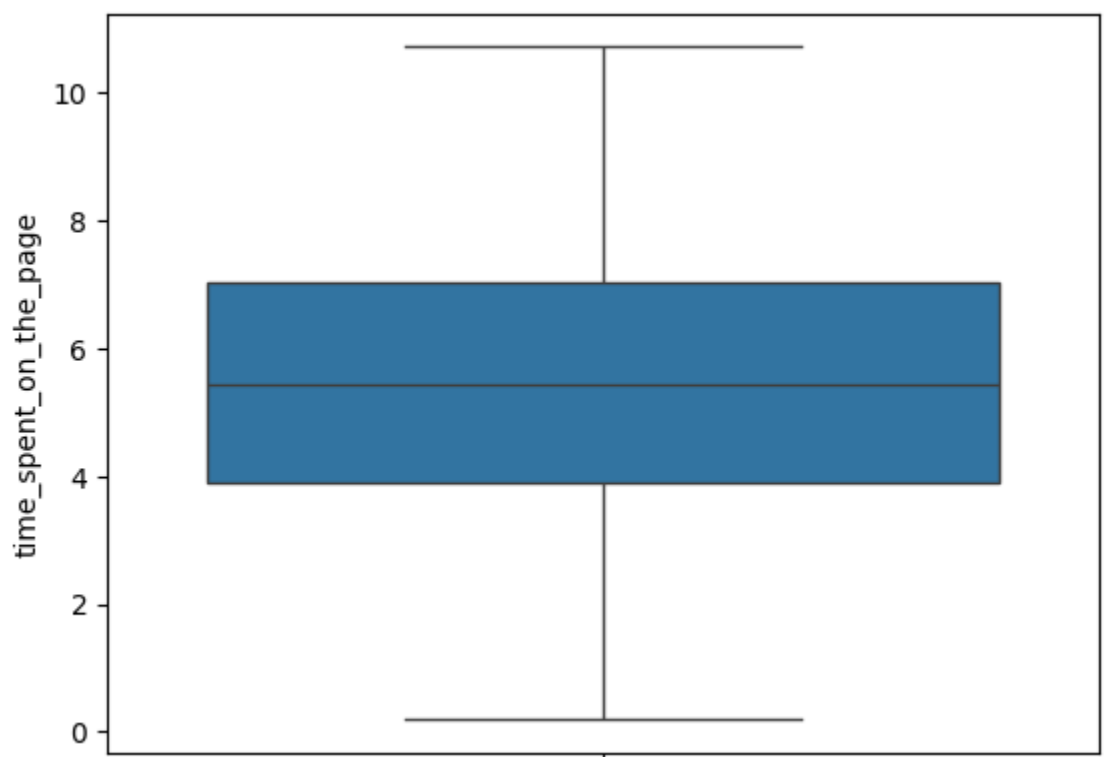
```
In [15]: # english has less users than that of spanish and french but not a significant amount
sns.histplot(df['language_preferred'])
```

```
Out[15]: <Axes: xlabel='language_preferred', ylabel='Count'>
```



```
In [16]: #mean time spent on page is around 5 min, min is 19 sec and max is around 11 min
sns.boxplot(df['time_spent_on_the_page'])
```

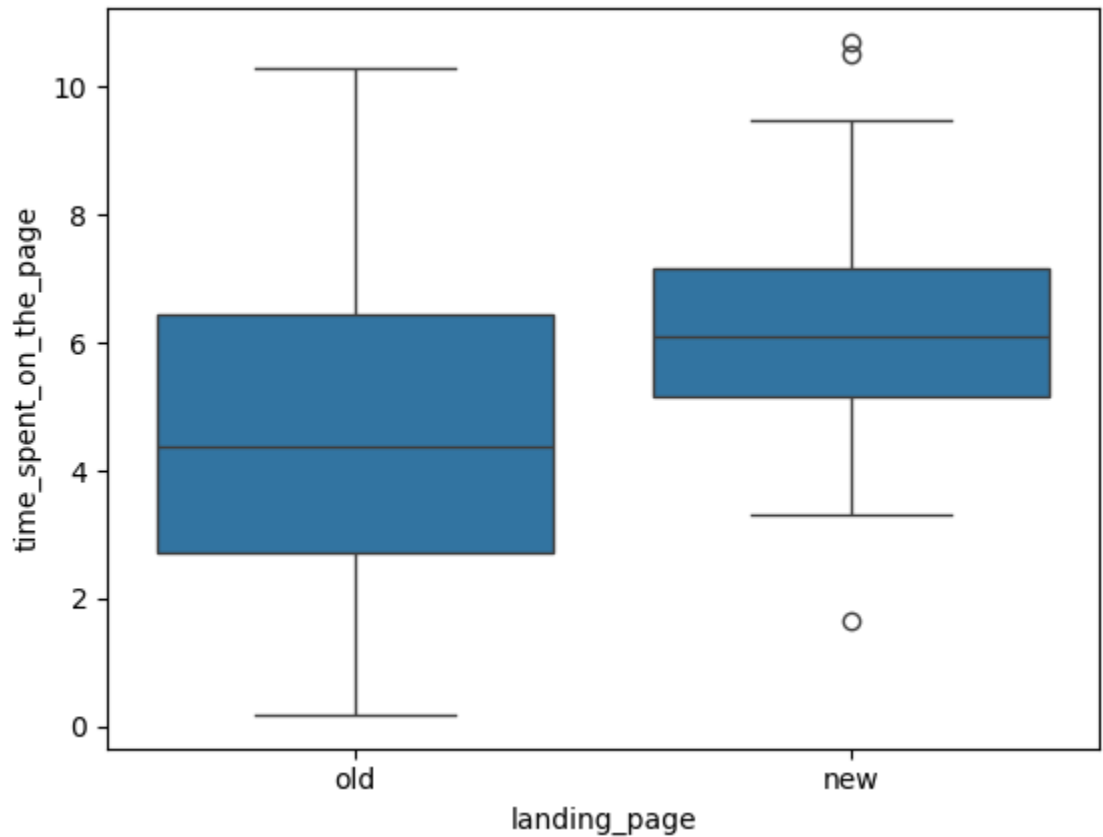
Out[16]: <Axes: ylabel='time_spent_on_the_page'>



Bivariate Analysis

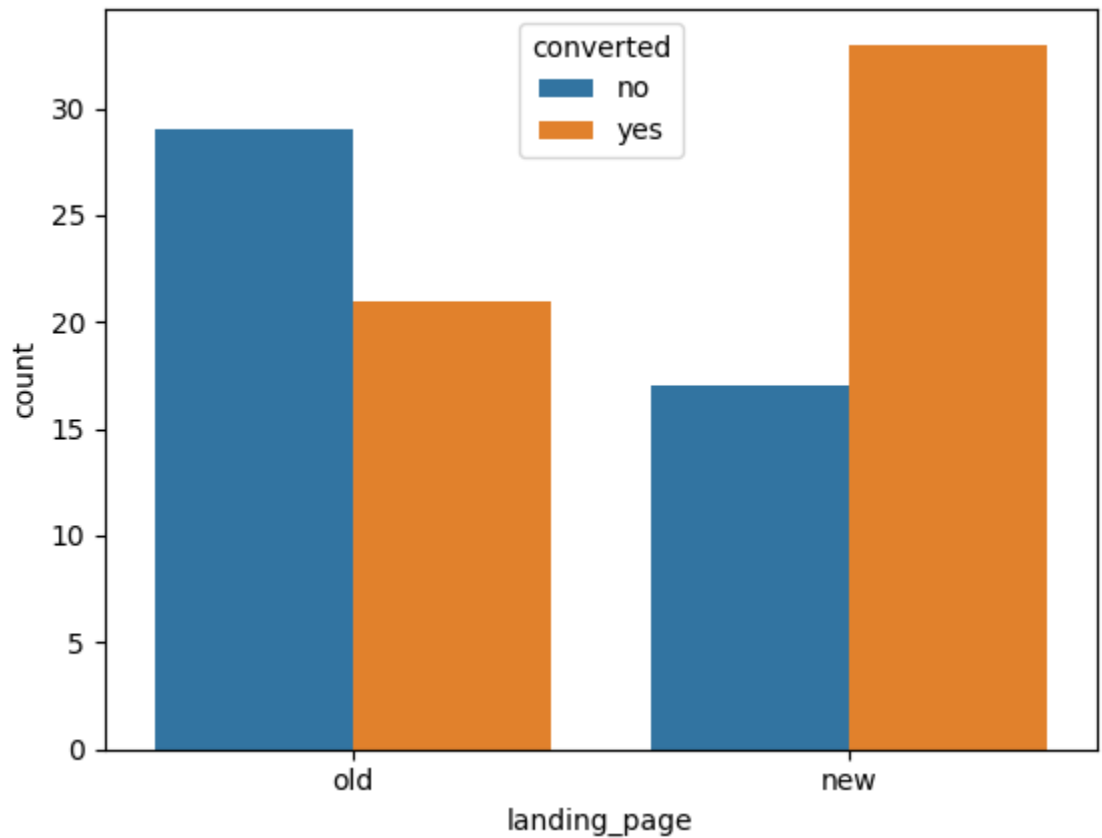
```
In [17]: #people are spending more time on the newer page overall
sns.boxplot(data = df,x = 'landing_page', y = 'time_spent_on_the_page')
```

Out[17]: <Axes: xlabel='landing_page', ylabel='time_spent_on_the_page'>



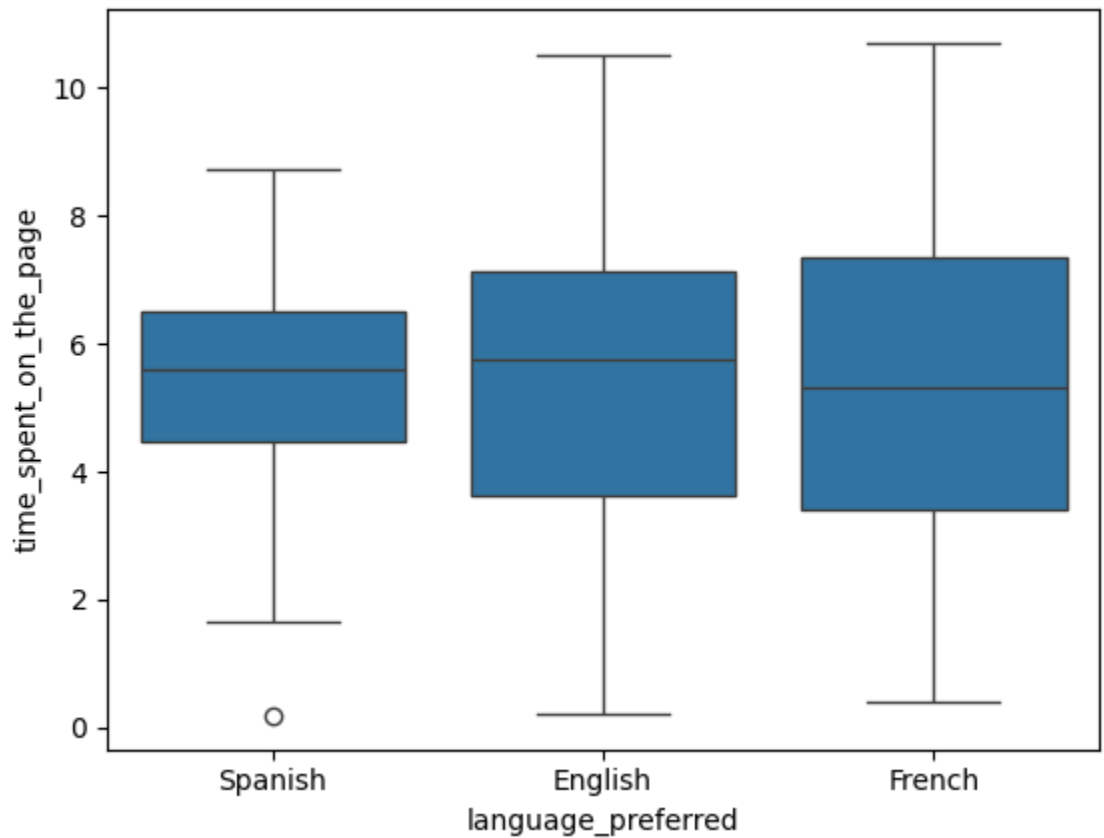
```
In [18]: #newer landing page has a higher conversion than the old landing page
sns.countplot(data = df,x = 'landing_page', hue = 'converted')
```

Out[18]: <Axes: xlabel='landing_page', ylabel='count'>



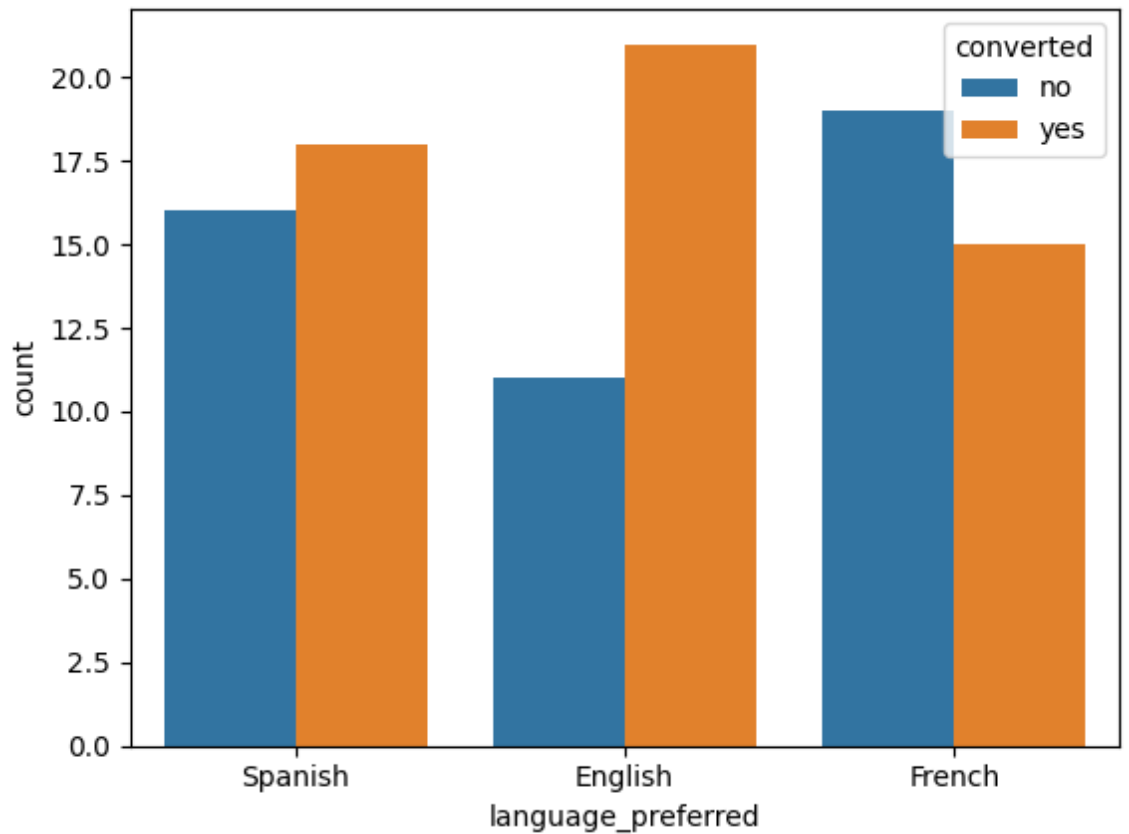
```
In [19]: #french had the highest time spent on page, while english has the lowest  
sns.boxplot(data = df, x = 'language_preferred', y = 'time_spent_on_the_page')
```

```
Out[19]: <Axes: xlabel='language_preferred', ylabel='time_spent_on_the_page'>
```



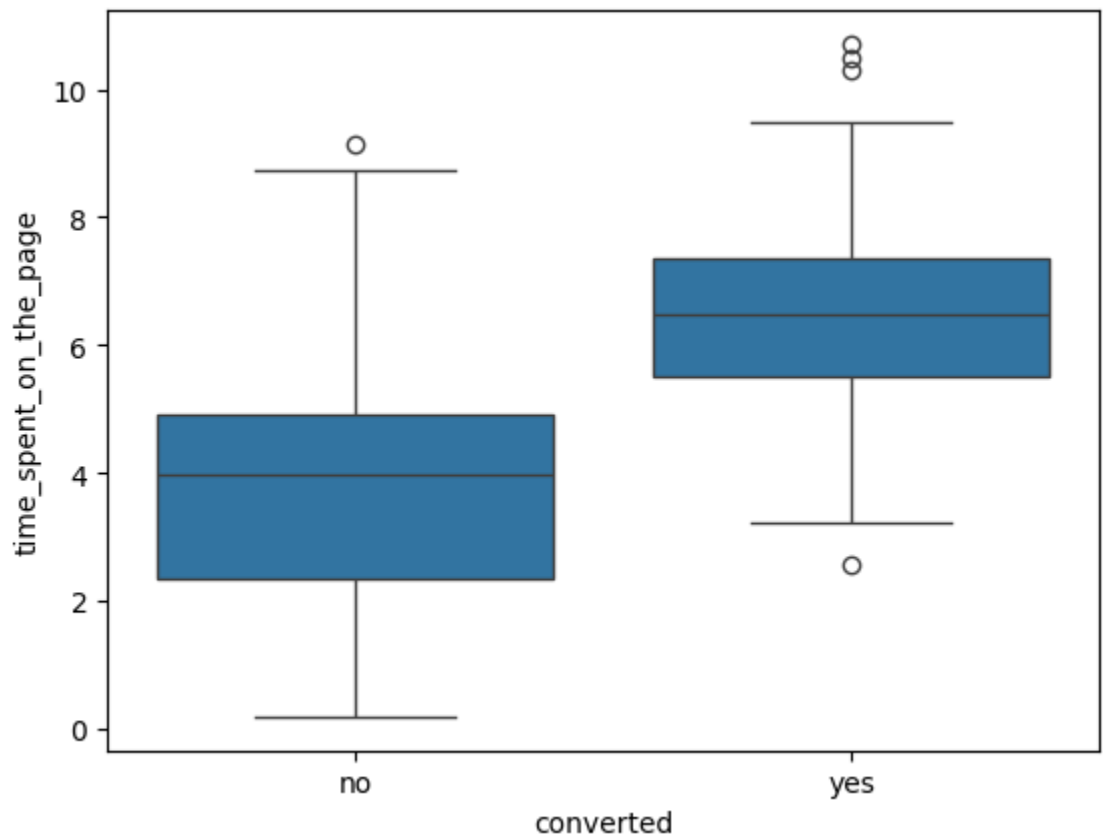
```
In [20]: # english users have the highest conversion, while french have the lowest
sns.countplot(data = df, x = 'language_preferred', hue = 'converted')
```

```
Out[20]: <Axes: xlabel='language_preferred', ylabel='count'>
```



```
In [21]: # spending more time on the page does convert them
sns.boxplot(data = df, x = 'converted', y = 'time_spent_on_the_page')
```

```
Out[21]: <Axes: xlabel='converted', ylabel='time_spent_on_the_page'>
```

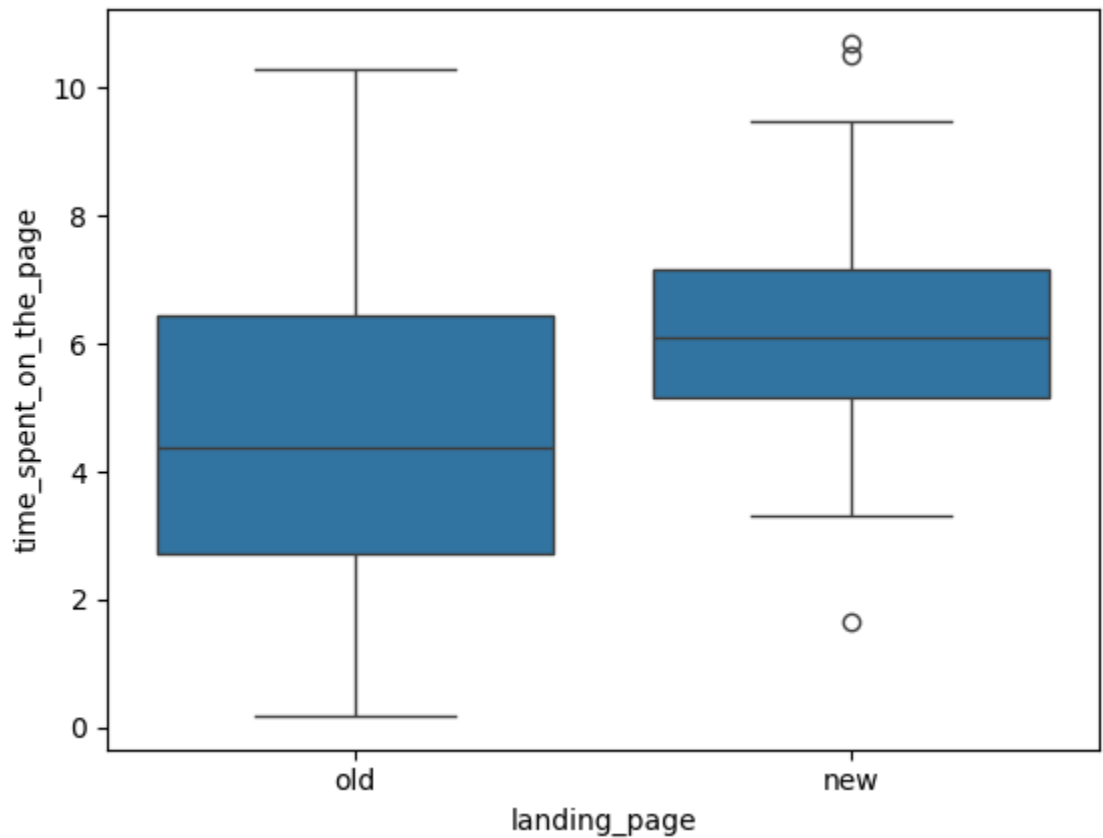


1. Do the users spend more time on the new landing page than the existing landing page?

Perform Visual Analysis

```
In [22]: sns.boxplot(data = df, x = 'landing_page', y = 'time_spent_on_the_page')
```

```
Out[22]: <Axes: xlabel='landing_page', ylabel='time_spent_on_the_page'>
```



Step 1: Define the null and alternate hypotheses

Null Hypothesis: $\mu_{old} \geq \mu_{new}$

Alternate Hypothesis: $\mu_{old} < \mu_{new}$

Step 2: Select Appropriate test

2 sample t test

Step 3: Decide the significance level

Use a significance level of 0.05

Step 4: Collect and prepare data

```
In [23]: #finding the standard deviations of time spent on the page on the new and old
landing page
new = df[df['landing_page'] == 'new']['time_spent_on_the_page']
old = df[df['landing_page'] == 'old']['time_spent_on_the_page']
print(round(new.std(),2))
print(round(old.std(),2))
```

1.82
2.58

Step 5: Calculate the p-value

```
In [24]: from scipy.stats import ttest_ind

test_stat, p_value = ttest_ind(new,old,equal_var=False,alternative='greater')
print(test_stat, p_value)
```

3.7867702694199856 0.0001392381225166549

Step 6: Compare the p-value with α

```
In [25]: if p_value < 0.05:
print(f'The p-value {p_value} is less than the level of significance, we
reject the null hypothesis.')
else:
print(f'The p-value {p_value} is greater than the level of significance,
we accept the null hypothesis.')
```

The p-value 0.0001392381225166549 is less than the level of significance,
we reject the null hypothesis.

Step 7: Draw inference

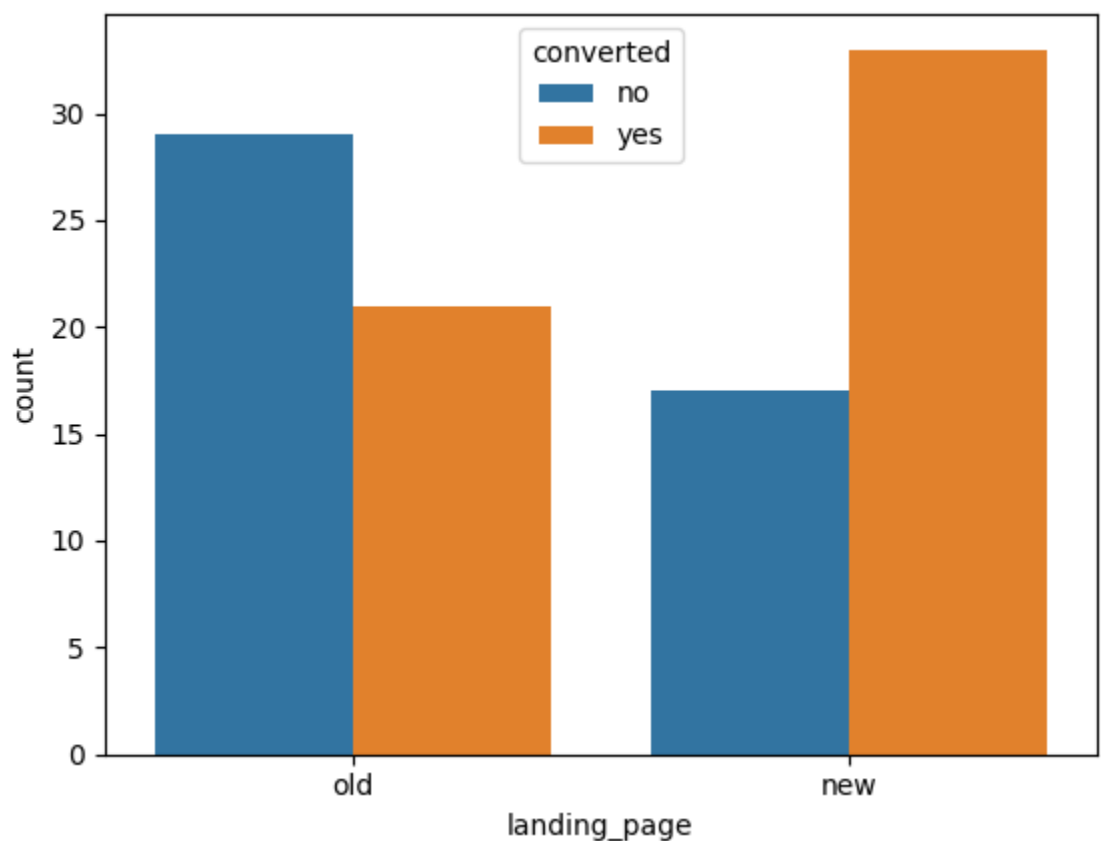
Based on the p value it is most likely that users spend more time on the newer landing page compared to the old one.

A similar approach can be followed to answer the other questions.

2. Is the conversion rate (the proportion of users who visit the landing page and get converted) for the new page greater than the conversion rate for the old page?

```
In [26]: sns.countplot(data=df, x='landing_page', hue='converted')
```

```
Out[26]: <Axes: xlabel='landing_page', ylabel='count'>
```



Null Hypothesis: $\mu_{\text{old}} > \mu_{\text{new}}$

Alternate Hypothesis: $\mu_{\text{old}} < \mu_{\text{new}}$

2 proportions z test

Significance level is still 0.05


```
In [27]: #Counting the yes of new and old landing page conversions and finding the
total num of users on the new and old page
new1 = df[df['landing_page'] == 'new']['converted'].value_counts()['yes']
old2 = df[df['landing_page'] == 'old']['converted'].value_counts()['yes']
numnew = df.landing_page.value_counts()['new']
numold = df.landing_page.value_counts()['old']
numnew, numold
```

```
Out[27]: (np.int64(50), np.int64(50))
```

```
In [28]: from statsmodels.stats.proportion import proportions_ztest

test_stat, p_value = proportions_ztest([new1, old2] , [numnew, numold],
alternative = 'larger')

test_stat, p_value
```

```
Out[28]: (np.float64(2.4077170617153842), np.float64(0.008026308204056278))
```

```
In [29]: if p_value < 0.05:
print(f'The p-value {p_value} is less than the level of significance, we
reject the null hypothesis.')
else:
print(f'The p-value {p_value} is greater than the level of significance,
we accept the null hypothesis.')
```

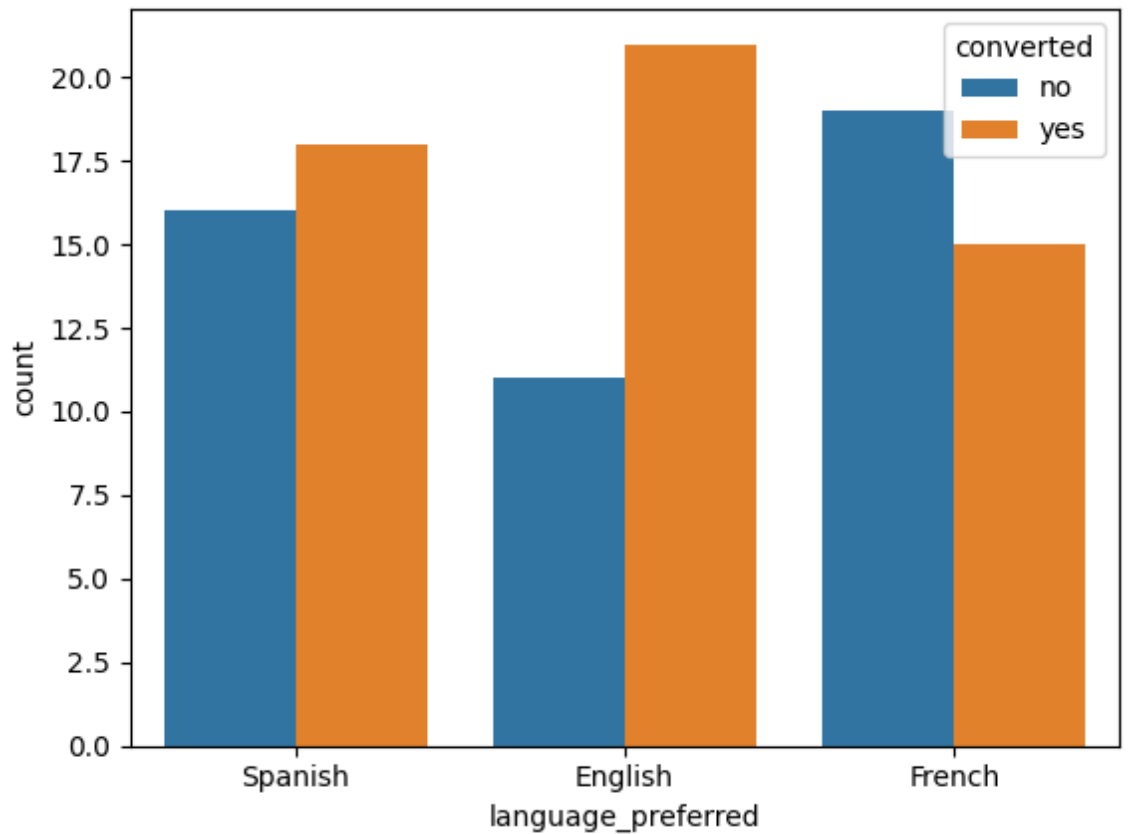
```
The p-value 0.008026308204056278 is less than the level of significance,
we reject the null hypothesis.
```

Since the p value is less than the level of significance and reject the null hypothesis it is most likely that the conversion rate of the new page is greater than the conversion rate of the old page

3. Is the conversion and preferred language are independent or related?

```
In [30]: sns.countplot(data = df, x = 'language_preferred', hue = 'converted')
```

```
Out[30]: <Axes: xlabel='language_preferred', ylabel='count'>
```



Null Hypothesis: conversion and preferred language are independent

Alternate Hypothesis: conversion and preferred language are dependent on each other

Chi Square test is going to be used because of two categorical variables

Significance level is 0.05

```
In [31]: #create contingency table
contingency_table = pd.crosstab(df['language_preferred'], df['converted'])
contingency_table
```

Out[31]:

	converted	
	no	yes
language_preferred		
English	11	21
French	19	15
Spanish	16	18

```
In [32]: from scipy.stats import chi2_contingency
chi2, p_value, dof, exp_freq = chi2_contingency(contingency_table)
p_value
```

```
Out[32]: np.float64(0.2129888748754345)
```

```
In [33]: if p_value < 0.05:
        print(f'The p-value {p_value} is less than the level of significance, we
        reject the null hypothesis.')
    else:
        print(f'The p-value {p_value} is greater than the level of significance,
        we accept the null hypothesis.')
```

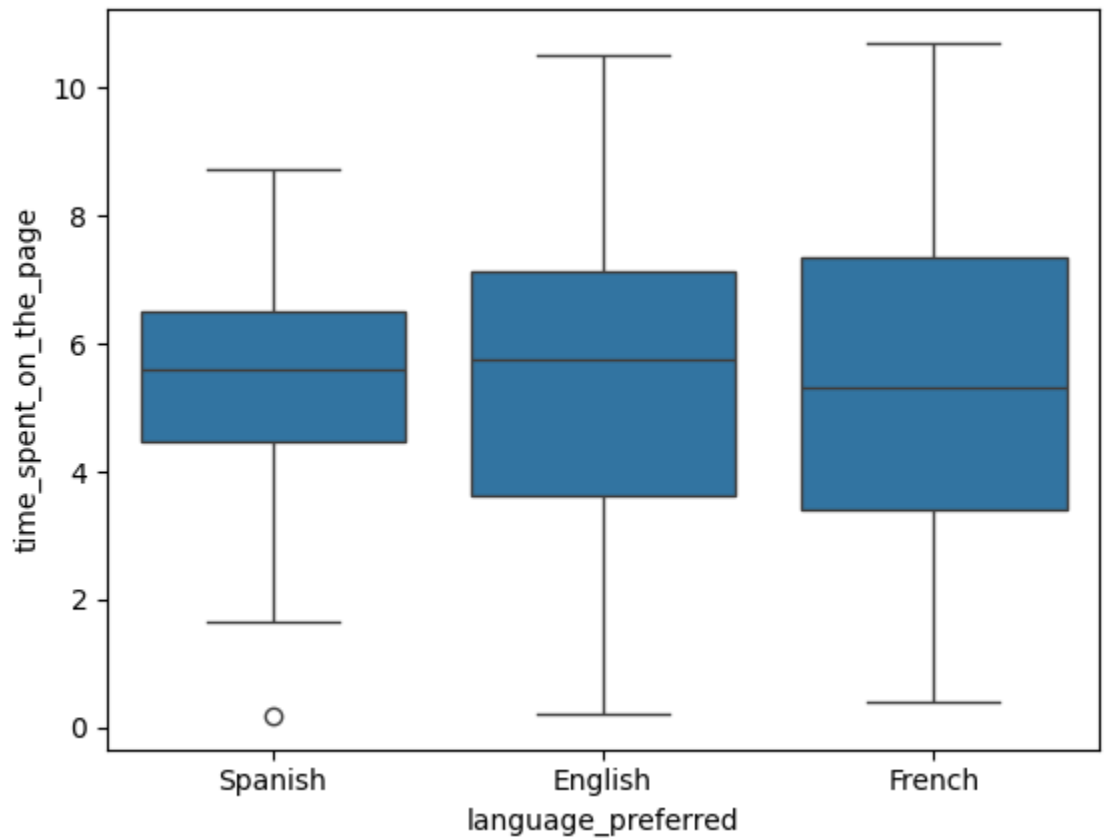
The p-value 0.2129888748754345 is greater than the level of significance, we accept the null hypothesis.

So since the null hypothesis is accepted because of the level of significance being less than the p value that means that conversion and language preferred are independent of each other.

4. Is the time spent on the new page same for the different language users?

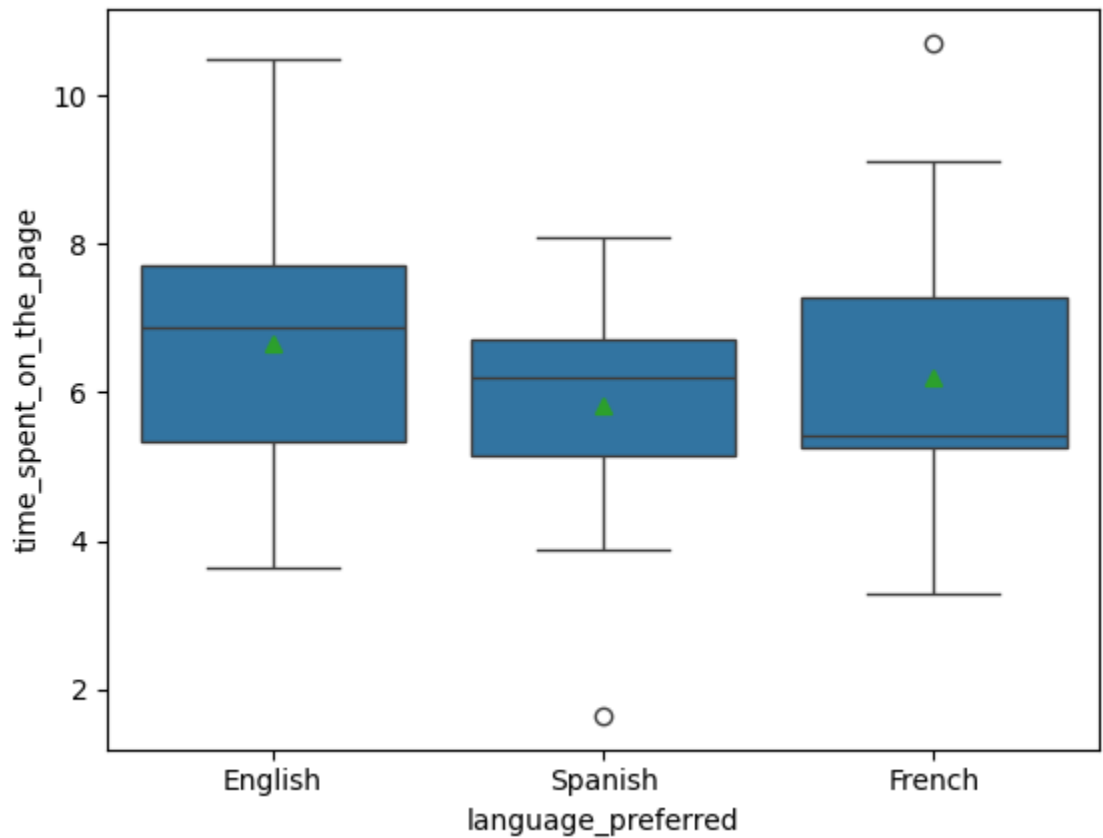
```
In [34]: sns.boxplot(data = df, x = 'language_preferred', y = 'time_spent_on_the_page')
```

```
Out[34]: <Axes: xlabel='language_preferred', ylabel='time_spent_on_the_page'>
```



```
In [35]: df_new = df[df['landing_page'] == 'new']
```

```
In [36]: sns.boxplot(x = 'language_preferred', y = 'time_spent_on_the_page', showmeans = True, data = df_new)
plt.show()
```



```
In [37]: df_new.groupby(['language_preferred'])['time_spent_on_the_page'].mean()
```

Out[37]:

	time_spent_on_the_page
language_preferred	
English	6.663750
French	6.196471
Spanish	5.835294

dtype: float64

Null Hypothesis: Time spent on new page is same for different language users

Alternate Hypothesis: Time spent on new page is different for different language users

One-way Anova F-test being used because of three means

Significance level is 0.05

```
In [38]: english = df_new[df_new['language_preferred']=="English"]
        ['time_spent_on_the_page']
        french = df_new[df_new['language_preferred']=="French"]
        ['time_spent_on_the_page']
        spanish = df_new[df_new['language_preferred']=="Spanish"]
        ['time_spent_on_the_page']
```

```
In [39]: from scipy.stats import f_oneway
        test_stat, p_value = f_oneway(english, french, spanish)
        p_value
```

```
Out[39]: np.float64(0.43204138694325955)
```

```
In [40]: if p_value < 0.05:
        print(f'The p-value {p_value} is less than the level of significance, we
        reject the null hypothesis.')
        else:
            print(f'The p-value {p_value} is greater than the level of significance,
            we accept the null hypothesis.')
```

The p-value 0.43204138694325955 is greater than the level of significance, we accept the null hypothesis.

So since the null hypothesis is accepted because of the level of significance being less than the p value that means that the time spent on the new page is the same across all of the languages preferred.

Conclusion and Business Recommendations

Business Recommendations

- * Deploy the new landing page since it is doing well on all fronts
- * Keep support for languages and maybe even add more since they do not negatively affect interaction
- * Try to streamline service so conversion rate from old to new is even higher
- * Add in user recommendations to bring more people to the site